

Shop-A-Lytics

GitHub repository checklist: what must be in each folder and file

Use this to confirm the repository is complete before the 3 Nov submission. Tick each box when the file exists and holds what the second column says. Owners are taken from the PM's task list; items marked "proposed" were unassigned and need confirming.

Status key

- EXISTS: already in the repository. EXISTS: UPDATE: in the repository but needs changes.
- ADD: still to create. GENERATED: made by a script or a build; listed in .gitignore, never committed. OPTIONAL: create only if useful.
- Design rule behind the layout: BoschCard data and the client's extra tables live in two separate databases. Only the files in backend/app/data/ read BoschCard data, so a changed dataset means editing column_settings.json or one module, not the whole app.

1. Repository root

✓	File / folder	What it must hold	Its function	Owner / status
☐	README.md	Project overview; setup from a fresh clone; how to build the two databases; how to run backend and frontend; how to use a different dataset; how to run tests; demo accounts; privacy mechanisms; known limitations; team. No TODO markers left.	The brief's required README: a stranger to the stack must be able to run the system from it alone.	Chris (outline), Nkazi (data sections) EXISTS: UPDATE
☐	.gitignore	node_modules/, frontend/dist/, .venv/, __pycache__/, .pytest_cache/, *.sqlite, .env and .env.*, data/prepared/, analytics_salt.txt, backend/reports/	Keeps secrets, databases and generated files out of Git.	Chris EXISTS: UPDATE
☐	.env.example	Placeholder values only: FLASK_APP, FLASK_DEBUG, SECRET_KEY, DATABASE_URL, ANALYTICS_DB_PATH, BOSCHCARD_SALT, MIN_GROUP_SIZE (10), MIN_HISTORY_DAYS (90), each with a comment.	Template each member copies to a private .env. Never holds real secrets.	Chris / Nkazi EXISTS
☐	package.json, package-lock.json	JavaScript dependencies and exact versions (React, Vite, Recharts).	Reproducible frontend installs. Confirm with Chris whether npm install runs at the root or in frontend/.	Chris EXISTS
☐	.env	Your real local settings and secrets.	Private to each person. Must never be committed.	Everyone <i>GENERATED (do not commit)</i>

2. docs/

✓	File / folder	What it must hold	Its function	Owner / status
☐	BRD_v1.2.pdf	The BRD corrected to the real stack (Flask, React, SQLite instead of Django, PostgreSQL), final scope and non-functional requirements.	Agreed reference between client and developers.	Nkazi ADD

✓	File / folder	What it must hold	Its function	Owner / status
☐	adaptation.md	How each client requirement was adapted: accepted, changed or declined, with reasons. States that the BoschCard data model is authoritative and the client's extra tables live in a separate app database.	Required deliverable inside the repo.	Nkazi ADD
☐	decisions.md	Decision log per session: the client ERD gaps (gps type, role values, feedback category, first-login password flag, audit log, calendar tables), the options, what was agreed and the date.	Evidence of structured elicitation; the one-page list taken to Session 3 (6 Oct).	Nkazi ADD
☐	data_requirements.md	Data requirements for the SA&D package: tables and fields used, volumes, history length, planted test cases, edge cases the data must cover.	The brief requires data requirements to be specified in the SA&D package.	Nkazi ADD
☐	data_dictionary.md	Column definitions, accounting conventions (value is gross, VAT included; discount is a rand amount; refunds are negative rows) and the seeded test features.	Explains the synthetic data to the team and the lecturers.	Nkazi ADD
☐	README_data_swap.md	Runbook for swapping in the lecturer's data: steps, common mismatches and fixes, how to practise.	Makes the 1 Nov data swap a 15-minute job.	Nkazi ADD
☐	stack_justification.md	Why Flask, React, SQLite, pandas, fpdf2 and pytest: team skills, fit for purpose, ecosystem maturity, feasibility, alternatives considered, and that all are free or open source.	The brief requires the stack choice to be justified in writing.	Chris / Nkazi ADD
☐	erd/	The client's ERD image and your own diagram showing the two databases (analytics and app) and the single vendor_id link.	Shows how the client ERD and the BoschCard model fit together.	Nkazi ADD
☐	meetings/	One note per client session (Sessions 1–6): purpose, questions asked, answers, decisions, outstanding items.	Records the elicitation process, which is assessed.	Nkazi / Iman ADD

3. data_tools/

Run from the repo root, for example: `python data_tools/validate_data.py data/synthetic`

✓	File / folder	What it must hold	Its function	Owner / status
☐	generate_boschcard_data.py	Seeded generator for the four BoschCard tables. Options: number of transactions and students, a real vendor list, -- anomalies for planted unusual days.	Produces reproducible synthetic data, and bigger or special datasets for testing.	Nkazi ADD
☐	validate_data.py	Checks any dataset against the ERD: files, columns, keys, foreign keys, date format, GPS order, discount ≤ value, duplicates. Prints ERROR / WARNING / INFO and exits non-zero on errors.	First line of defence against a bad dataset.	Nkazi ADD
☐	prepare_data.py	Maps a supplied dataset onto the standard ERD using column_settings.json, then validates it and writes validation_report.txt.	Absorbs differences in the lecturer's data without code changes.	Nkazi ADD

✓	File / folder	What it must hold	Its function	Owner / status
☐	load_analytics.py	Builds the privacy-safe analytics.sqlite: no names, emails, phones, addresses or birth dates; salted student pseudonym; age band; money in cents; split latitude and longitude; date, hour, weekday and month columns.	The only way BoschCard data enters the application.	Nkazi ADD
☐	load_data.py	Plain copy of the four CSVs into SQLite, personal fields included.	Reference and debugging only. The application must never read it.	Nkazi ADD
☐	column_settings.json	File and column names, datetime and birth-date formats, GPS order, money unit, discount type, value type, salt variable, age bands, residence pattern.	The one file edited on 1 Nov when the lecturer's data arrives.	Nkazi ADD
☐	vendor_list_template.csv	Header and example rows: name, address, gps, type, popularity, mode.	Format for loading a real vendor list into the generator.	Nkazi ADD
☐	test_etl.py	Twelve pytest tests: no personal data in the analytics file, stable and salted hashes, money totals match, lecturer-style data maps back, clear errors on bad data, planted anomalies present.	Proves the data layer is safe and swap-ready. Run: <code>pytest -q data_tools/test_etl.py</code>	Nkazi ADD

4. data/

✓	File / folder	What it must hold	Its function	Owner / status
☐	synthetic/student.csv, vendor_type.csv, vendor.csv, transaction.csv	The default 60,000-transaction dataset in the BoschCard format (about 4 MB).	Lets anyone run the system straight after cloning.	Nkazi ADD
☐	synthetic/ground_truth_promotions.csv	The 24 promotions injected into the data.	Test-only answer key for the “association, not causation” feature. The application must never read it.	Nkazi ADD
☐	reference/public_holidays.csv	South African public holidays 2025–2026 (date, name, note). Verify against gov.za.	Calendar context and the “explained by calendar” rule for anomalies.	Nkazi ADD
☐	reference/university_calendar.csv	Columns start_date, end_date, type (term, exam, recess, event), name, source. Currently header only: fill in from the university's published calendar.	Feeds calendar shading and the university context requirement (RC3).	Nkazi ADD
☐	analytics.sqlite	Privacy-safe BoschCard data, built by load_analytics.py.	Read-only data source for the application; rebuilt when new data arrives.	Script <i>GENERATED (do not commit)</i>
☐	app.sqlite	Users, feedback, reports, audit log, calendar events.	Application data; a data reload never touches it (NFR 9).	Script <i>GENERATED (do not commit)</i>

✓	File / folder	What it must hold	Its function	Owner / status
☐	prepared/	Standardised CSVs and validation_report.txt from prepare_data.py.	Intermediate output; read the report after every load.	Script <i>GENERATED (do not commit)</i>
☐	synthetic_anomalies/	Larger dataset (200,000 rows, about 10 MB) with planted spikes and outages, plus ground_truth_anomalies.csv.	Test data for anomaly detection. Generate on demand; do not commit.	Nkazi <i>OPTIONAL</i>

5. scripts/

✓	File / folder	What it must hold	Its function	Owner / status
☐	scripts/build_databases.py	One command that: validates the data; builds analytics.sqlite; creates the app tables; seeds demo owner and member accounts linked to the vendors actually loaded; checks for users whose vendor no longer exists; prints the demo logins.	Single reproducible setup step for the README, and the re-run after the lecturer's data arrives.	Nkazi (proposed) ADD

6. backend/

6.1 Backend root

✓	File / folder	What it must hold	Its function	Owner / status
☐	backend/run.py	Entry point that starts the Flask app.	Run the backend: python run.py	Chris EXISTS
☐	backend/pyproject.toml, requirements.txt	Python project settings; dependencies: Flask, SQLAlchemy, pandas, numpy, fpdf2, pytest.	Reproducible backend installs.	Chris EXISTS: UPDATE
☐	backend/reports/	PDF and CSV exports created by the app.	Kept outside any publicly served folder; not committed.	Iman <i>GENERATED (do not commit)</i>

6.2 backend/app/

✓	File / folder	What it must hold	Its function	Owner / status
☐	app/__init__.py	App factory: creates the Flask app, loads config, opens both databases, registers routes, runs the readiness check at start-up.	Wires the whole backend together.	Chris EXISTS
☐	app/extensions.py	Shared objects such as the SQLAlchemy instance.	Avoids circular imports.	Chris EXISTS
☐	app/security.py	Password hashing and checking (Werkzeug), session helpers, owner-only and login-required decorators.	Login and role-based access (RC6, NFR 4).	Iman / Chris EXISTS

✓	File / folder	What it must hold	Its function	Owner / status
☐	app/config.py	Reads .env: database paths, secret key, MIN_GROUP_SIZE, MIN_HISTORY_DAYS.	Keeps thresholds out of the code.	Chris ADD
☐	app/models/app_models.py	App-database tables only: User (vendor_id, email, password_hash, full_name, role, must_change_password), Report, Feedback (with category), AuditLog, CalendarEvent. No student data.	The client's extra entities, separate from BoschCard data (NFR 9).	Nkazi + Chris (proposed) ADD
☐	app/data/analytics.py	Read-only connection to analytics.sqlite and every query function (spend per vendor by day, week, month; breakdowns; location search). Always filtered to the logged-in user's vendor.	The only code that reads BoschCard data. If the data deviates, change this and column_settings.json only.	Chris ADD
☐	app/data/readiness.py	Start-up check: row counts, date span, vendors per type, history per vendor.	Turns short or thin data into "Not assessed" and "not enough data" screens instead of crashes (NFR 8, NFR 10).	Nkazi + Chris ADD
☐	app/data/privacy.py	Minimum group size (10) and suppression helpers.	Hides small groups so individuals cannot be identified (NFR 2, NFR 3).	Iman ADD
☐	app/services/anomaly.py	Anomaly rules, plain-language explanations, calendar-explained exclusions, "Not assessed: insufficient history".	RC7, NFR 6, NFR 10. Zero or far-below-normal days need an explicit rule.	Chris ADD
☐	app/services/baselines.py	Historical baselines for comparing promotion periods; wording that says "observed", never "caused".	RC8, NFR 7.	Chris / Bou ADD
☐	app/services/pdf_export.py	fpdf2 report builder and CSV export.	Report export (RC4); records each export in Report.	Iman ADD

6.3 backend/app/routes/

✓	File / folder	What it must hold	Its function	Owner / status
☐	routes/auth.py	Sign up, log in, log out, first-login password change.	Accounts (RC6).	Iman ADD
☐	routes/team.py	Owner-only: list, add and remove team members.	Owner vs Member roles.	Iman ADD
☐	routes/kpis.py	Headline metrics endpoint.	Dashboard KPIs (RC2).	Bou ADD
☐	routes/spend.py	Spend over time with month → week → day drill-down.	RC1.	Bou ADD
☐	routes/breakdowns.py	Breakdowns by day of week, hour, discount and so on.	RC2.	Bou ADD
☐	routes/alerts.py	Anomaly flags and "Not assessed" states.	RC7.	Chris ADD

✓	File / folder	What it must hold	Its function	Owner / status
☐	routes/search.py	Location and vendor search.	RC9.	Chris ADD
☐	routes/feedback.py	Feedback form endpoint with category (promotion insights vs usability).	RC5.	Nkazi ADD
☐	routes/reports.py	PDF and CSV export endpoints.	RC4.	Iman ADD

6.4 backend/tests/

✓	File / folder	What it must hold	Its function	Owner / status
☐	tests/conftest.py, test_health.py	Test setup and the health-check test.	Existing skeleton.	Chris EXISTS
☐	tests/test_privacy.py	Fails if any response contains a student identifier or a group below MIN_GROUP_SIZE.	Concrete evidence for NFR 2 and NFR 3.	Iman ADD
☐	tests/test_vendor_isolation.py	Fails if one vendor's user can reach another vendor's data.	Concrete evidence for NFR 4.	Chris / Bou ADD
☐	tests/test_readiness.py	Short, empty and thin datasets give "Not assessed" or "not enough data", not errors.	NFR 8 and NFR 10.	Nkazi ADD
☐	tests/test_datasets.py	The app starts and key pages respond on four datasets: default, planted anomalies, lecturer-style, tiny short-history.	Protects the 1 Nov swap.	Nkazi / Bou ADD

7. frontend/

✓	File / folder	What it must hold	Its function	Owner / status
☐	index.html, package.json, vite.config.js	Page shell, frontend dependencies, Vite and proxy settings.	Frontend build and dev server.	Chris EXISTS
☐	src/main.jsx, App.jsx, styles.css	Entry point, routing and layout, global styles.	Frontend skeleton.	Chris EXISTS
☐	dist/	Built frontend.	Build output; ignored by Git.	Script <i>GENERATED (do not commit)</i>
☐	src/api/client.js	The single place that calls the backend; handles errors and logged-out responses.	Keeps API calls out of components.	Chris ADD
☐	src/pages/Login.jsx, SetPassword.jsx, Team.jsx	Login, first-login password change, team management.	Accounts and roles.	Iman ADD

✓	File / folder	What it must hold	Its function	Owner / status
☐	src/pages/Dashboard.jsx, Detail.jsx	Main dashboard; drill-down detail view with breadcrumb.	RC1, RC2.	Bou ADD
☐	src/pages/Alerts.jsx	List of anomaly flags with plain-language explanations.	RC7.	Chris ADD
☐	src/pages/Feedback.jsx	Categorised feedback form.	RC5.	Nkazi ADD
☐	src/components/SpendChart.jsx	Recharts chart with chart-type switcher and compare.	Core visualisation.	Bou ADD
☐	src/components/CalendarShading.jsx	Shaded bands for terms, exams, recesses and holidays on charts.	University context (RC3).	Nkazi ADD
☐	src/components/NotEnoughData.jsx	The “not enough data” screen shown when a segment is too small or history too short.	Privacy and NFR 10.	Nkazi ADD
☐	src/components/ExportDialog.jsx	Dialog for choosing report type and downloading.	RC4.	Iman ADD

8. Never commit

What	Why
.env, analytics_salt.txt	Secrets. A leaked salt lets pseudonyms be reversed by guessing student numbers.
*.sqlite, data/prepared/	Generated databases and intermediates; rebuild them with the scripts.
node_modules/, frontend/dist/, .venv/, __pycache__/, .pytest_cache/	Dependencies and build output.
backend/reports/	Generated PDF exports.
The lecturer's dataset	Confirm with the lecturers before committing it. Commit only the synthetic data.

9. Before you submit (due Tue 3 Nov, 23:59)

✓	Check	Why it matters
☐	Fresh-clone test: clone into a new folder or another laptop, follow the README only, and confirm the app runs.	The brief: “a stranger to your stack should be able to run your system from these instructions alone.”
☐	Swap rehearsal: run build_databases.py on a lecturer-style dataset at least once before 1 Nov.	The real data arrives about two days before the deadline.
☐	All tests pass: pytest -q data_tools/test_etl.py and pytest -q in backend/.	Evidence of quality and privacy mechanisms.

✓	Check	Why it matters
☐	README has no TODO markers; demo accounts are listed.	Required deliverable.
☐	docs/adaptation.md complete and BRD_v1.2 in docs/.	Required deliverable (Tue 3 Nov, 23:59).
☐	git ls-files shows no .env, .sqlite or analytics_salt.txt.	Secrets and generated files stay out.
☐	Every member has regular commits throughout the build.	The brief requires professional collaboration.
☐	Git bundle created as the lecturers instruct and uploaded to SocSciLearn before the deadline.	No late submissions are accepted.
☐	Backup screen recording of the demo recorded.	Safety net for the presentation (5/6 Nov).
☐	Each member's SCARF self-evaluation submitted separately.	Due Tue 3 Nov, 23:59.

10. Brief requirements and where the repository meets them

Brief requirement	Where it is met	Evidence
Working system	backend/, frontend/, scripts/build_databases.py	App runs end to end on synthetic and swapped data
Requirements management	docs/decisions.md, docs/adaptation.md, docs/meetings/	Agreed scope with the client, recorded per session
Persistence	data/analytics.sqlite and data/app.sqlite via SQLAlchemy	Two SQLite databases
Free but justified stack	docs/stack_justification.md	Written justification
Realistic synthetic data	data_tools/generate_boschcard_data.py, data/synthetic/, docs/data_dictionary.md	Seeded, validated, matches the data model
Privacy, sensitivity and responsible analytics	load_analytics.py, app/data/privacy.py, vendor isolation tests, audit log, anomaly and baseline wording	Tests that fail if a mechanism is bypassed
Reproducible setup	README.md, .env.example, scripts/build_databases.py	Fresh-clone test passes
Professional collaboration	Git history	Regular commits by all four members